

DEPARTMENT OF
COMPUTER SCIENCE B.Sc. Part - III Computer Science (Entire)
Shivraj College Of Arts, Commerce & D.S.Kadam Science College
Gadhinglaj 2023-2024

SEMINAR REPORT ON

BIOLOGICAL INSPIRATION

By

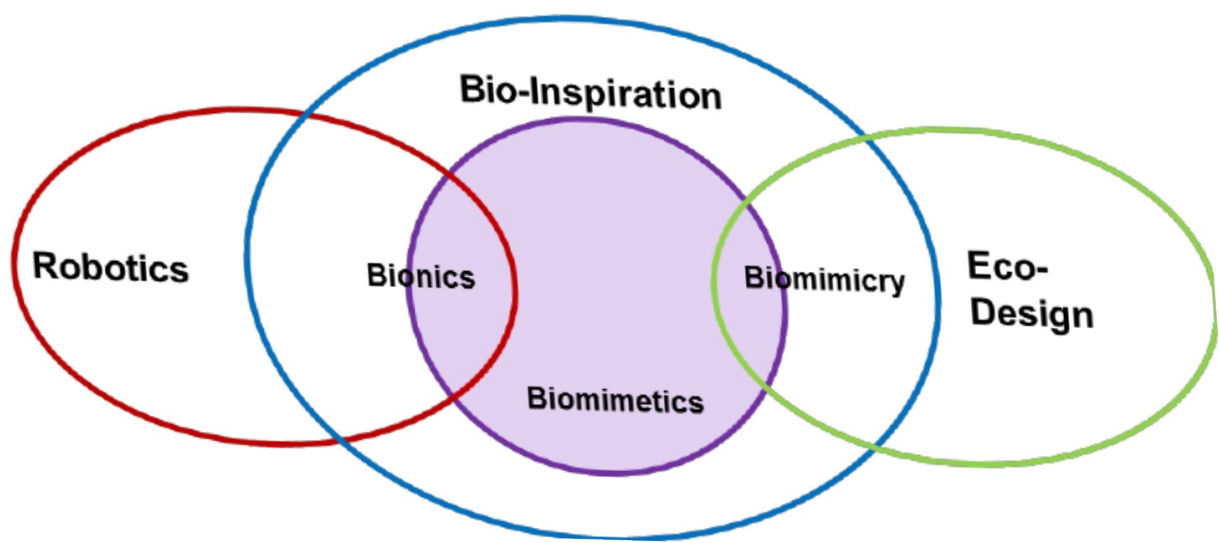
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➤ INTRODUCTION

Biological materials are ingeniously designed and optimized tools that are employed by nature for organisms to survive and thrive within challenging environments. They represent the elegant strategies that fulfill a variety of not only mechanical but also functional needs as they are ^{biomimicry} generally simple in composition but efficient in performance. This is distinct from most materials that usually depend on complex chemicals or expensive manufacturing, and therefore often confront a trade off between properties (e.g., increasing weight to increase strength). Thus, biological materials have been an endless source of inspiration for developing novel materials and structures in recent decades. To actuate this inspiration, the first fundamental step requires revealing structure-property mechanisms and formulating systematic theories, which is known as Biological Materials Science. This paves the road for the next exciting step of creating new advanced materials by providing essential insights with heretofore unexploited strategies from natural designs.



➤ **Development of bio inspired materials**

biological and bio inspired materials by focusing on two categories: functional and structural, both of which depend on their structures. Within each category, we will focus on particularly impactful and recent examples of bio inspired designs, and therefore provide a compact overview of the grand panorama of this field. Bio inspired functional materials will focus on designs that provide super wet ability, bioactivity, and stimuli-responsiveness, while bio inspired structural materials will include designs that provide high strength with low weight and high toughness with low weight.

➤ **Characteristic of biological inspiration**

1 Biologically-inspired design is based on analogies requiring expertise across two disparate domains (e.g. architecture and biology), and thus is inherently interdisciplinary.

2 Since the objects, relations, and processes across domains are very different, design collaborators speak from different lexicons.

3 Since biologists in general seek to understand the functions of designs occurring in nature whereas designers generally seek to generate designs for human needs, they use different methods of investigation and have different perspectives on design.

4 The resources, such as materials and processes, available in nature to realize an abstract design concept are very different from the resources available in the human domain.

➤ **Types of Biological Inspiration**

1 **Bio utilization:** This type of biological inspiration involves using living organisms or their by products to create new products or technologies

2 **Biomorphism:** This type of biological inspiration involves taking inspiration from the shapes, patterns, and color found in nature to create new designs

3 **Bio mimicry:** This type of biological inspiration involves copying the form and function of living organisms to create new products or technologies.

➤ **ADVANTAGES**

1 Biological control can be cost-effective in the long run. Although it may cost a bit to introduce a new species to an environment, it's a tactic that only needs to be applied once due to its self-perpetuating nature.

2 Most important of all, it's effective. Whatever pest population you want controlled will no doubt be controlled. Because the predator introduced will be naturally inclined to target the pests, very often you'll see the pest population dwindle.

➤ **DISADVANTAGES**

1 If you're looking to completely wipe out a pest, biological control is not the right choice. Predators can only survive if there is something to eat, so destroying their food population would risk their own safety. Therefore, they can only reduce the number of harmful pests.

2 While it is cheap in the long run, the process of actually setting up a biological control system is a costly endeavor. A lot of planning and money goes into developing a successful system.